INCH-POUND

MIL-C-71128 (AR)
AMENDMENT 2
31 May 1996

SUPERSEDING
AMENDMENT 1
09 November 1994

MILITARY SPECIFICATION

CARTRIDGE, 81MM PRACTICE, SHORT RANGE
M880 LOADING, ASSEMBLING AND PACKING

This amendment forms a part of MIL-C-71128 (AR), dated 26 May 1992, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

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3.3.7 Add the following:

3.3.7 Dud plugs.

3.3.7.1 Configuration. Three dud plugs shall be present and properly inserted into the projectile body.

* 3.3.7.2 Dud plug push in - push out force. No dud plug shall push into the required test plate using a force greater than 40.0 pounds and shall not be capable of being pushed out of the required test plate using a force of less than 25.0 pounds at a temperature of 70 degrees F ± -3 degrees F when tested as specified in 4.5.5."

3.4.1: Delete in its entirety and substitute:

"3.4.1 Misfires. The cartridges shall not fail to fire from the weapon when they are initially dropped into it."

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3.4.4: Delete in its entirety and substitute:

* "3.4.4 Range and deflection precision. The cartridge shall have a horizontal range probable error (PEr) less than 15 meters for all zones when calculated to the nearest one tenth (0.1) meter. The deflection probable error (PEd) shall be less than 15 meters for all zones when calculated to the nearest one tenth (0.1) meter)."

Add: "3.4.9 Dud plug function.

* 3.4.9.1 Functioned round. All dud plug heads (regardless of stems) shall blow out of the projectile body upon fuze function."

* 3.4.9.2 Inert round. No dud plug head (regardless of stems) shall exit the projectile body during ballistic flight or upon impact."

4.3.3: Delete "g" and "h" in their entirety and substitute:

"g. If the weapon is damaged by abnormal wear of the interior or of the firing pin, or caused by excessive pressure.

h. Two or more improper impact signatures occur."

Add the following:

"i. All three dud plug heads (regardless of stems) are missing from the projectile body in the inert fuze ballistic test.

j. Two or more dud plug heads (regardless of stems) fail to exit projectile bodies (accumulatively, i.e., two remaining in a single projectile or one each remaining in two projectiles) in the live fuze ballistic tests (except in the case of a particular round that was an actual dud, in which case paragraph "i." above applies)."

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* Table I, Cartridge, 81mm, Practice Short Range, under Conformance Criteria:

Delete: "72", and substitute: "96".

* Add the following:

"Dud Plug ...(dwg. 12961001)

Examination for defects ...30 ... 3.1 ... 4.4.2.9

Push in - push out test ... 30 ... 3.3.7.2 ... 4.5.5".

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4.4.1: Delete the word "interfix" from the text from lines a.,b., c., d, and e.

Add the following:

* "f. Dud plugs from one inspection lot."

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4.4.2.1, Major 105: Change call out:

from: "Length of large outside diameter section",

to: "Length between large outside diameter aft face to firing pin point".

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4.4.2.5, under Special: Add:

"H Any dud plug(s) missing ... 100% ... 3.3.7 ... Visual".

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* 4.4.2.9: Add the following:

PARAGRAPH	TITLE			DRAWING NUMBER
"4.4.2.9	Dud Plug	SHEET 1 OF 1		12961001
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Push in - push out test	30	3.3.7.2	Test/4.5.5
Minor				
201	Poor workmanship	30	3.6	Visual"

4.4.3.1.1, Add:

"i. All dud plug heads out of any one projectile in the inert fuze ballistic test, Critical defect (see 3.4.9.2).".

j. Two or more dud plug heads remain in the live fuze ballistic test, Major defect (see 3.4.9.1).".

4.4.3.1.2: Delete "i" and "j" in their entirety and substitute:

"i. All three dud plug heads (regardless of stems) are missing from the projectile body in the inert fuze ballistic test.

j. Two or more dud plug heads (regardless of stems) fail to exit projectile bodies (accumulatively, i.e., two remaining in a single projectile or one each remaining in two projectiles) in the live fuze ballistic tests (except in the case of a particular round that was an actual dud, in which case paragraph "i." above applies)."

4.5.1.1: Add the following:

"d. One group of 24 each with inert fuzes conditioned for at least 24 hours at 0 degrees F ± 2.5 degrees F (see 4.4.3.1.2), shall be fired at charge 3."

* 4.5.1.1, last subparagraph: Delete and substitute:

"The individual deflections and ranges (accuracy of both to 0.01 meter) and muzzle velocities shall be calculated to the nearest 0.1 meter and recorded on the firing record for groups a, b, and c above."

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Add the following:

* "4.5.5: Push in - push out test. Major defect. A hardened steel test plate of 1 x 2 x 1/4 inch is required for this test. There is to be a 0.250 ± .001 diameter thru hole (having an internal surface finish of 32 maximum) in the steel plate. The sample of dud plugs to be tested shall be preconditioned at a temperature of 100 degrees F ± 5 degrees F for a period of not less than 24 hours nor more than 48 hours prior to being subjected to this test. The preconditioned dud plugs shall be tested not earlier than 15 minutes after removal from the conditioning box nor later than one (1) hour after removal from the conditioning box. Press a sample dud plug into the hole using an arbor press until the head is 0.0625 ± 0.02 inch above flush with the plate surface. At this point, the push in test is completed. Record the maximum force. Turn the steel plate over and push the dud plug out of the steel plate to 0.0625 ± 0.02 inch above flush using the same arbor press. Record the maximum force required to push out the dud plug. At this point, the push out test is completed. Compute sample mean (xbar) and standard deviation (sd). Insert these into the following formulas:

Accept if:

a. xbar(In) = 37.0 pounds max., xbar(In) + 3*sd(In) = 40.0 pounds max.

and

b. xbar(Out) = 30.0 pounds min., xbar(Out) - 3*sd(Out) = 25.0 pounds min. and no individual x(Out) less than 25.0 pounds.

Any dud plug sample that does not meet the above requirements shall be cause for lot rejection.

Each test sample quantity may be examined for outliers using the Grubbs T statistical test (see 6.12). A maximum of two outliers may be removed from the sample quantity."

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* 6.10 Delete in its entirety.

6.11 Add the following:

"6.11 Inspection lot definition. A homogeneous collection of units of product or one shipment of component parts or assemblies from a subcontractor, bearing identification and treated as a unique entity from which a sample is to be drawn and inspected (and tested) to determine conformance with the acceptability criteria."

* 6.12: Add the following:

"6.12 Outlier test. When the accept criteria is not met and if it is suspected that a statistical outlier may be present in the sample of 30 observations. Arrange the observed values in order of increasing magnitude (X1, X2 .. Xn-1, Xn. Using the statistics calculated for the sample, calculate Tn where:

$$Tn = \frac{Xn - \overline{X}}{s}$$

If X1, the smallest value, rather than Xn, is the doubtful value the test criterion is:

$$T1 = \frac{\overline{X} - X1}{s}$$

Where:

$\overline{X}$ = arithmetic average of all n values
s = estimate of the population standard deviation based on the sample data calculated as follows:

$$s = \left[\frac{\sum (x_i - \overline{x})^{1/2}}{n-1}\right] = \left[\frac{n\sum x_i^2 - \left(\sum x_i\right)^2}{n(n-1)}\right]^{1/2}$$

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If Tn is less than or equal to 2.745, the highest value may not be eliminated. The lot is rejected. If Tn is greater than or equal to 2.745, the largest value obtained in the sample shall be eliminated and the sample statistics X and s shall be recalculated using the twenty-nine remaining values. This shall be compared to the accept criteria. If the criteria is met, accept the lot. If not, the same procedure shall be followed for the Xn-1 observation. Calculate Tn-1 based on the new statistics of the remaining sample of 29. If Tn-1 is less than or equal to 2.730, the lot is rejected. If Tn-1 is greater than or equal to 2.730, then this observation shall be eliminated and the sample statistics calculated based on the remaining 28 observations. If these values meet the accept criteria, accept the lot. If the accept criteria is still not met, reject the lot."

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